

Network Layer

In this layer, there are multiple other smaller layers:

- Hops: the number of router-to-router steps a packet takes to reach its destination.
- Routing protocols: rules that help routers choose the best path for data to travel across networks.
 - RIP (Routing Information Protocol): a simple protocol that picks routes by the fewest hops; best for small networks.
 - OSPF (Open Shortest Path First): a faster, smarter protocol that learns the network map and picks efficient routes inside an organization.
 - BGP (Border Gateway Protocol): the protocol that connects different networks on the internet and decides how traffic moves between ISPs and large providers.
- All of these use a routing table, which is like a router's address book: it tells the router where to send a packet next so it can reach its destination.
- The main protocol here is the IP protocol (Internet Protocol): it provides logical addressing and packet delivery across interconnected networks.
 - IP is connectionless, so each packet is routed independently and may take a different path.
 - It works with addresses like IPv4 and IPv6 to identify the source and destination of data.
 - Routers read the destination IP address and forward the packet toward the next hop using routing tables.
 - IP does not guarantee delivery, ordering, or error recovery; those are handled by higher-layer protocols.
 - It also supports fragmentation and reassembly when packets are too large for a network link.
- Errors, and signaling is done through the ICMP protocol:
 - ICMP (Internet Control Message Protocol) is used by routers and hosts to report network problems and diagnostic information.
 - It does not carry user data like TCP or UDP; instead, it sends control messages such as destination unreachable, time exceeded, and redirect.
 - Tools like ping and traceroute rely on ICMP to test reachability and measure the path packets take across the network.

Main functions of the network layer:

- Data plane: handles the actual movement of packets from one router to the next.
- Control plane: decides the best path packets should take across the network.
- Forwarding: sends each packet to the correct next hop using the routing table.
- Routing: builds and updates the paths that packets should follow to reach their destination.

Main SLAs of the Network layer is guaranteed delivery, delivery with bounded delay and in-order packet delivery, guaranteed minimal bandwidth, guaranteed minimal jitter, and other security services (CIA triad)

Network Services Type

- Connection-oriented service: a path is established before data is sent.
 - The sender and receiver set up a session first.
 - Packets usually follow the same route, so delivery is more predictable and often in order.
 - It adds setup overhead, but reliability and consistency are better.
- Connectionless service: each packet is sent independently without setup.
 - No session is created before transmission.
 - Packets may take different paths and can arrive out of order.

- It is simpler and faster to start, but less predictable.
- Virtual circuits: a connection-oriented service where a path is established before data is sent.
 - Packets usually follow the same route.
 - The network can provide more predictable delivery and ordering.
 - Examples include ATM and frame relay style networks.
- Datagram network: a connectionless service where each packet is routed independently.
 - No setup is needed before sending data.
 - Packets can take different paths and may arrive out of order.
 - The network stays simpler and more flexible, which is why IP uses this model.

IP Addresses

- An IP address is just a numerical label associated with each device connected to a computer network, which uses the IP protocol for communication.
- Two main functions:
 - Network/host identification: the IP address is split into a network part and a host part, so routers can tell which network a device belongs to and which specific device to reach.
 - Routing and delivery: routers use the network portion to forward packets toward the correct network, then the host portion to deliver the packet to the final device inside that network.
 - The prefix length (like /24 or /16) tells us how many bits belong to the network part.
 - This is why two devices can share the same network prefix but still have different host addresses.
 - In practice, the IP address acts like a location label, helping data move across interconnected networks without needing to know the device's physical hardware address.
- IP address space is managed in a hierarchy:
 - IANA (Internet Assigned Numbers Authority) manages the global pool of IP addresses and delegates large blocks to regional registries.
 - RIRs (Regional Internet Registries) distribute address blocks to countries, ISPs, universities, and organizations in their regions.
- The main Regional Internet Registries are:
 - ARIN (American Registry for Internet Numbers)
 - RIPE NCC (RIPE Network Coordination Centre)
 - APNIC (Asia-Pacific Network Information Centre)
 - LACNIC (Latin America and Caribbean Network Information Centre)
 - AFRINIC (African Network Information Centre)
- This system prevents conflicts and helps keep IP allocation coordinated across the internet.
- Organizations usually do not receive random individual IPs from IANA; they get address blocks from an ISP or RIR, and then subnet them internally.
- Some address ranges are reserved for special use and are not publicly routed, such as private networks, loopback, and link-local addresses.

IPv4 addresses were distributed by IANA to the RIRs in blocks of approximately 16.8 million addresses.

- These public IPv4 blocks were eventually exhausted at the IANA level.
- DMUC uses the private Class A 10.0.0.0/8 network.

IPv4 Rundown

IPv4 is the version of IP most people still recognize today. It uses a 32-bit address, meaning there are 2^{32} possible combinations in total, which is about 4.3 billion addresses. In practice, some of these are reserved for special uses, so the number of publicly routable addresses is smaller. IPv4 addresses are usually written as four decimal numbers separated by dots, like 192.168.1.10.

- Each IPv4 address has two parts:
 - Network part: tells routers which network the device is on.
 - Host part: identifies the device inside that network.
- A prefix like /24 tells us how many bits belong to the network part.
- Routers use the network part to move packets between networks.
- Once the packet reaches the right network, the host part gets it to the correct device.
- IPv4 has a limited number of addresses, so private ranges, subnetting, and NAT became important.

192	168	1	10
Network	Network	Host	Host

Table 1: A simple IPv4 address split into network and host parts.

- Common private IPv4 ranges:
 - 10.0.0.0/8
 - 172.16.0.0/12
 - 192.168.0.0/16
- Special-purpose addresses:
 - 127.0.0.1 for loopback
 - 169.254.0.0/16 for link-local
 - 255.255.255.255 for broadcast

IPv4 Classes and /8

- IPv4 used to be divided into classes, called classful addressing.
- The class was determined by the first bits of the address, which also changed the default prefix:
 - Class A: /8
 - Class B: /16
 - Class C: /24
 - Class D: multicast
- A /8 prefix means the first 8 bits are the network part and the remaining 24 bits are the host part.
- That gives a Class A network a very large address space: $2^{24} - 2 = 16,777,214$ usable host addresses.
- 10.0.0.0/8 is a private Class A block, so any address from 10.0.0.0 to 10.255.255.255 stays inside private networks unless routed specially.
- Classful addressing is mostly historical now, but it was the main reason IPv4 space ran out so quickly. Under classful rules, organizations got fixed-size blocks: Class A (/8), Class B (/16), or Class C (/24).
- This caused massive waste. A company that needed a few thousand addresses might receive a whole Class B block with 65,534 usable hosts, while a much larger organization might still outgrow what it had. Because blocks were handed out in large, rigid chunks, huge portions of the IPv4 space were allocated long before they were actually used.
- The exhaustion happened in stages:
 - First, the usable public IPv4 pool kept shrinking as more devices, ISPs, universities, and companies connected to the internet.

- Second, early allocation policies were generous and inefficient, so many organizations held address blocks much larger than they needed.
- Third, the internet exploded in size, especially with home networks, mobile devices, cloud services, and always-connected systems.
- Finally, IANA ran out of free public IPv4 space and could no longer hand out large new blocks to the RIRs; after that, the regional registries also became tight on supply.
- The main solutions were:
 - CIDR (Classless Inter-Domain Routing): replaced fixed classes with variable-length prefixes like /12, /20, and /27, so networks could get blocks that matched their actual size needs.
 - Route aggregation: CIDR also let routers summarize many smaller networks into one advertisement, reducing routing-table growth.
 - Subnetting: organizations split one allocated block into smaller internal networks instead of requesting many separate public blocks.
 - NAT (Network Address Translation) and PAT (Port Address Translation): allowed many private devices to share one public IPv4 address.
 - Private address ranges: 10.0.0.0/8, 172.16.0.0/12, and 192.168.0.0/16 were reserved for internal use so organizations could reuse the same space locally without consuming public IPv4.
 - IPv6: introduced a much larger 128-bit address space to solve the long-term shortage.
- How they were implemented:
 - CIDR changed how addresses are assigned and advertised. Instead of assuming the class from the first bits, routers and administrators use the prefix length explicitly, like 203.0.113.0/27.
 - NAT is usually implemented at the network edge, such as a home router or ISP gateway. Internal devices use private IPs, and the NAT device rewrites source addresses and ports so many hosts can appear as one public IP.
 - IPv6 is deployed with dual-stack systems, tunneling, and translation where needed, so networks can gradually move from IPv4 without breaking old services.
 - RIR policies became stricter, so new IPv4 allocations are smaller and based on real need rather than historical class sizes.
- In short, IPv4 was exhausted because it was distributed inefficiently, fixed-size classes wasted space, and internet growth outpaced the 32-bit address pool. CIDR, NAT, private addressing, and IPv6 are the main responses to that shortage.
- Even though classes are obsolete in practice, they are still useful for understanding older documents, default masks, and why /8, /16, and /24 are such common shorthand.

CIDR and Subnet Masking

- CIDR (Classless Inter-Domain Routing) is a way to write an IP network with a prefix, like 192.168.1.0/24.
- The /24 part is the subnet mask in prefix form. It tells how many bits are used for the network.
- A bigger prefix number means a smaller network:
 - /24 = 256 addresses total, 254 usable
 - /16 = much larger network
- Subnet masking is the process of separating the network part from the host part of an IP address.
- Routers use the mask to know whether a packet stays local or gets forwarded.
- In short: CIDR is the notation, and the subnet mask defines the network size.

IPv6 Rundown

- IPv6 is the newer version of IP designed to replace IPv4 and solve address exhaustion.
- It uses 128 bits, which gives an enormous address space: 2^{128} possible addresses.

- An IPv6 address is written in hexadecimal, grouped into 8 blocks of 16 bits, like 2001:db8:abcd:12::1.
- Its main composition is:
 - Global routing prefix: identifies the larger network on the internet.
 - Subnet ID: identifies the local subnet inside that network.
 - Interface ID: identifies the device or interface on that subnet.
- IPv6 supports features like:
 - Stateless Address Autoconfiguration (SLAAC) for automatic address setup.
 - DHCPv6 for managed address assignment.
 - Link-local addresses for local network communication, usually starting with fe80::/10.
 - Multicast instead of broadcast; IPv6 does not use broadcast addresses.
- Common shorthand rules:
 - Leading zeros can be omitted.
 - One run of all-zero blocks can be compressed with :: only once per address.
- IPv6 is usually deployed with dual stack, where IPv4 and IPv6 run side by side during migration.
- Early IPv6 history and proposal work appear in documents such as RFC 1752, and later IPv6-era documents include RFC 2480 among many others.
- In short, IPv6 keeps the same basic job as IPv4, but with a much larger address space, cleaner autoconfiguration, and better long-term scalability.

Class	IP Range Starts With...	Default Prefix	Hosts per Network	Historical Use Case
Class A	1 to 126	/8	16,777,214	Governments, massive ISPs, global tech giants
Class B	128 to 191	/16	65,534	Mid-sized companies, universities
Class C	192 to 223	/24	254	Small businesses, home networks
Class D	224 to 239	N/A	None (Multicast)	One-to-many data streaming

- A router is a networking device that connects different networks and directs packets between them.
- Its two primary functions are:
 - Forwarding: sending each packet to the correct next hop using the destination IP address and routing table.
 - Routing: building and updating the paths in the routing table so packets can reach their destination.

RIP, OSPF, and BGP Comparison

- These three routing protocols solve different problems:
 - RIP is simple and works best in small networks.
 - OSPF is designed for fast, efficient routing inside an organization.
 - BGP connects different networks across the internet and follows routing policy.

Protocol	Type / Scope	How It Chooses a Path	Best For	Easy Summary
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RIP	Interior Gateway Protocol IGP used inside small networks	Chooses the route with the fewest hops. It does not look deeply at link speed or quality.	Small networks with simple topologies	Simple, easy to learn, but not very scalable
OSPF	Interior Gateway Protocol IGP used inside organizations and enterprises	Builds a full map of the network and uses shortest-path calculations based on link cost.	Medium to large internal networks	Faster and smarter than RIP for internal routing
BGP	Exterior Gateway Protocol EGP used between autonomous systems	Uses policy and path attributes such as AS-path, local preference, and next hop.	Internet-scale routing between ISPs and large organizations	The main protocol that keeps the internet connected

- Quick way to remember them:
 - RIP: fewest hops.
 - OSPF: best internal path.
 - BGP: best policy between networks.
- Main difference:
 - RIP and OSPF are mostly for routing **inside** one organization.
 - BGP is for routing **between** different organizations and networks.

Link Layer

- The link layer moves frames between neighboring nodes on the same link.
- It is responsible for local delivery, not end-to-end internet delivery.
- Examples:
 1. Ethernet.
 2. Wi-Fi.
 3. Point-to-point links.
- Important link-layer concepts:
 1. **Frame**: link-layer data unit.
 2. **MAC address**: hardware/link-layer address, such as 00:1a:a0:b4:4b:c1.
 3. **Network adapter/NIC**: hardware that sends and receives frames.
 4. **Error detection**: checking whether a frame was corrupted.

CSMA

- CSMA means Carrier Sense Multiple Access.
- It is a media-access idea used when multiple devices share the same transmission medium.
- A device listens before transmitting:
 1. If the medium is idle, it may transmit.
 2. If the medium is busy, it waits.
- Ethernet historically used CSMA/CD, where CD means collision detection.
- Wi-Fi uses CSMA/CA, where CA means collision avoidance, because wireless devices cannot reliably detect collisions in the same way as wired Ethernet.

ARP

- ARP maps an IPv4 address to a MAC address on the local network.

- A host uses ARP when it knows the destination IP but needs the destination MAC address for the Ethernet or Wi-Fi frame.
- Basic ARP flow:
 1. Host broadcasts: “Who has this IP address?”
 2. The matching host replies with its MAC address.
 3. The sender stores the result in its ARP cache/table.
- ARP is local to a subnet. Routers do not forward ARP broadcasts across normal routed networks.

Ethernet and Switching

- Ethernet frames contain source and destination MAC addresses.
- A switch learns which MAC addresses are reachable on which ports by observing incoming frames.
- If the switch knows the destination MAC, it forwards the frame only out the correct port.
- If it does not know the destination MAC, it floods the frame out other ports.
- Switches operate mainly at layer 2, while routers operate at layer 3.

CRC and Error Detection

- CRC means cyclic redundancy check.
- The sender computes a check value from the frame bits and includes it with the frame.
- The receiver recomputes the check value.
- If the values do not match, the receiver treats the frame as corrupted.
- CRC detects many common transmission errors, but it is not encryption and does not prove authenticity.

File Transfer and Remote Access

Netkit and Kathara

- Netkit is a network emulation tool used in older networking labs.
- It lets students run multiple virtual Linux machines connected by virtual links, so routing, addressing, and packet capture can be practised safely.
- This repo also contains Kathara lab material. Kathara is a more modern Docker-based replacement for many Netkit-style labs.
- Typical lab workflow:
 1. inspect the topology files.
 2. start the lab.
 3. run commands such as ip, ping, tcpdump, and route.
 4. inspect packets in Wireshark.
 5. stop and clean the lab.
- These labs help connect theory to real protocol behaviour.

FTP

- FTP means File Transfer Protocol.
- It shares files and directories between remote systems using a client-server model.
- FTP uses TCP and traditionally uses two connections:
 1. port 21 for commands/control.
 2. port 20 or a negotiated port for data.
- FTP can use different representations:
 1. ASCII mode for text.
 2. Image/binary mode for byte-for-byte transfer.
 3. EBCDIC mode for IBM text.

- 4. Local mode for compatible systems using a local format.
- Data transfer modes include:
 1. stream mode.
 2. block mode.
 3. compressed mode.

Active and Passive FTP

- In active mode, the client opens the control connection and the server opens a data connection back to the client.
- Active mode often fails through NAT or firewalls because the client may not accept incoming connections from the internet.
- In passive mode, the client asks the server for a data port and then the client opens the data connection.
- Passive mode is commonly used when clients are behind NAT or firewalls.

FTP Security

- Plain FTP can expose usernames, passwords, commands, and file contents.
- Common risks include:
 1. brute-force login attempts.
 2. packet sniffing.
 3. spoofing.
 4. bounce attacks.
 5. port stealing.
- Secure alternatives:
 1. **FTPS**: FTP extended with SSL/TLS.
 2. **SFTP**: SSH File Transfer Protocol. It is not the same protocol as FTP.
 3. **SCP**: secure copy over SSH.
 4. rsync over SSH for efficient synchronization.

SSH

- SSH means Secure Shell.
- It is a cryptographic protocol for secure remote login, remote command execution, tunneling, and file transfer.
- Default port: 22.
- Common commands:

```
ssh user@host
ssh -p 2222 user@host
sftp -P 2222 user@host
scp file.txt user@host:/path/
```

- SSH uses cryptography to authenticate the server and encrypt the session.
- Users may authenticate with passwords or public/private key pairs.

Web Security and TLS

- HTTPS is HTTP carried over TLS.
- TLS provides:
 1. confidentiality through encryption.
 2. integrity protection against undetected modification.
 3. server authentication using certificates.
- SSL is the older predecessor to TLS and is now considered deprecated.

Certificates and Certificate Authorities

- A certificate binds a public key to a domain name or identity.
- A certificate authority (CA) signs certificates so clients can decide whether to trust a server.
- Browsers and operating systems include trusted CA root certificates.
- If a server presents a valid certificate for the requested domain, the browser can verify that it is talking to the expected server.

TLS Handshake

- A simplified TLS handshake:
 1. Client offers supported TLS versions and cipher suites.
 2. Server chooses protocol parameters.
 3. Server sends its certificate.
 4. Client verifies the certificate.
 5. Client and server establish shared session keys.
 6. Later HTTP data is encrypted using fast symmetric encryption.
- Public-key cryptography is mainly used during setup and authentication.
- Symmetric encryption is used for the main data transfer because it is faster.

DHCP and the DORA Cycle

- DHCP means Dynamic Host Configuration Protocol.
- It automatically gives hosts network settings such as:
 1. IP address.
 2. subnet mask.
 3. default gateway.
 4. DNS server.
 5. lease time.
- DHCP uses UDP and is connectionless.
- DORA cycle:
 1. **Discover**: client broadcasts to find DHCP servers.
 2. **Offer**: server offers an available address and settings.
 3. **Request**: client requests one offered address.
 4. **Acknowledge**: server confirms the lease.
- Clients often broadcast because they do not yet have a valid IP address.
- DHCP relay is used when clients and DHCP servers are on different subnets.
- A client may use ARP to check whether an offered address is already in use.

DNS

- DNS means Domain Name System.
- It translates human-readable names into IP addresses, such as resolving a web domain to an IPv4 or IPv6 address.
- DNS is hierarchical:
 1. root servers.
 2. top-level domain servers, such as .com, .org, or country-code TLDs.
 3. authoritative name servers for specific domains.
 4. recursive resolvers used by clients.
- Common DNS record types:
 1. A: name to IPv4 address.
 2. AAAA: name to IPv6 address.
 3. CNAME: alias to another name.

4. MX: mail exchanger for email delivery.
 5. NS: authoritative name server.
 6. TXT: text records, often used for verification and email security.
- DNS responses are cached for their TTL, which reduces load and improves performance.

Recursive Lookup

- A typical DNS lookup:
 1. The client asks a recursive resolver.
 2. The resolver asks a root server where to find the TLD server.
 3. The resolver asks the TLD server where to find the domain's authoritative server.
 4. The resolver asks the authoritative server for the record.
 5. The resolver returns the answer to the client and caches it.

DNS Security Issues

- Traditional DNS responses are not encrypted and were not designed with strong authentication.
- DNS cache poisoning tries to insert false records into a resolver cache.
- Attackers may use misleading domains for phishing.
- DNSSEC adds signatures to DNS data so resolvers can validate authenticity and integrity, but it does not encrypt the query.

Quick Acronyms

Acronym	Meaning
ARP	Address Resolution Protocol
CSMA	Carrier Sense Multiple Access
CRC	Cyclic Redundancy Check
DHCP	Dynamic Host Configuration Protocol
DNS	Domain Name System
DORA	Discover, Offer, Request, Acknowledge
FTP	File Transfer Protocol
RFC	Request for Comments
SFTP	SSH File Transfer Protocol
SSH	Secure Shell
SSL	Secure Sockets Layer, older predecessor to TLS
TLS	Transport Layer Security
TLD	Top-Level Domain